ESGEM-AMR

ESCMID

AMRrules

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Agenda

1. AMRrules v1.0 release

2. AMRrules Specification v1.1 – proposed updates

3. Sourcing geno/pheno data

- AMRgen R package
- AllTheBacteria commandline tool

4. AST data import/export

- Vitek export instructions; volunteers for others?
- Web app for AST data conversion (using AMRgen)

4. Planning

- Manuscript plans, ESCMID 2027

AMRrules v1.0 1,272 rules defined in 32 species

Priorities

- ✓ Core genes in key AMR pathogens (ESKAPEE + WHO priority list)
- ✓ Phenotypes defined by EUCAST
- ✓ Compatible with AMRFinderPlus & NCBI databases

Browsing: Klebsiella pneumoniae

Displaying 7 row(s).

Download TSV | Columns to display

ruleID	organism	gene	mutation	variation type	gene context	drug	drug class	phenotype	clinical category	breakpoint standard	PMID	evidence gr...	evidence limitations
KPN0001	Klebsiella pneumoniae	blaSHV	-	Gene presence detected	core	-	penicillin beta-lactam	wildtype	R	EUCAST Expected resistant phenotypes v1.2 (2023)	32284385	high	-
KPN0002	Klebsiella pneumoniae	oqxA	-	Gene presence detected	core	ciprofloxacin	-	wildtype	S	EUCAST v15.0 (2025)	30834112	moderate	low clinical relevance, lacks evidence for this allele
KPN0003	Klebsiella pneumoniae	oqxB	-	Gene presence detected	core	ciprofloxacin	-	wildtype	S	EUCAST v15.0 (2025)	30834112	moderate	lacks evidence for this species
KPN0004	Klebsiella pneumoniae	fosA5_fam	-	Gene presence detected	core	fosfomycin	-	wildtype	S	ECOFF (May 2025)	33128341	moderate	lacks evidence for this species
KPN0005	Klebsiella	fosA5	-	Gene	core	fosfomycin	-	wildtype	S	ECOFF (May	25441705	moderate	lacks evidence for

- ✓ All organisms
- Acinetobacter baumannii
- Bordetella bronchiseptica
- Bordetella hinzii
- Bordetella holmesii
- Bordetella parapertussis
- Bordetella pertussis
- Burkholderia pseudomallei
- Campylobacter jejuni
- Enterobacter hormaechei
- Enterobacter roggenkampii
- Enterococcus faecalis
- Enterococcus faecium
- Escherichia coli
- Klebsiella grimontii
- Klebsiella huaxiensis
- Klebsiella michiganensis
- Klebsiella oxytoca
- Klebsiella pasteurii
- Klebsiella spallanzanii
- Klebsiella pneumoniae
- Legionella
- Legionella longbeachae
- Legionella pneumophila
- Mycobacterium tuberculosis
- Neisseria gonorrhoeae
- Neisseria meningitidis
- Proteus mirabilis
- Pseudomonas aeruginosa
- Salmonella enterica
- Shewanella
- Staphylococcus aureus
- Yersinia enterocolitica
- Yersinia pseudotuberculosis

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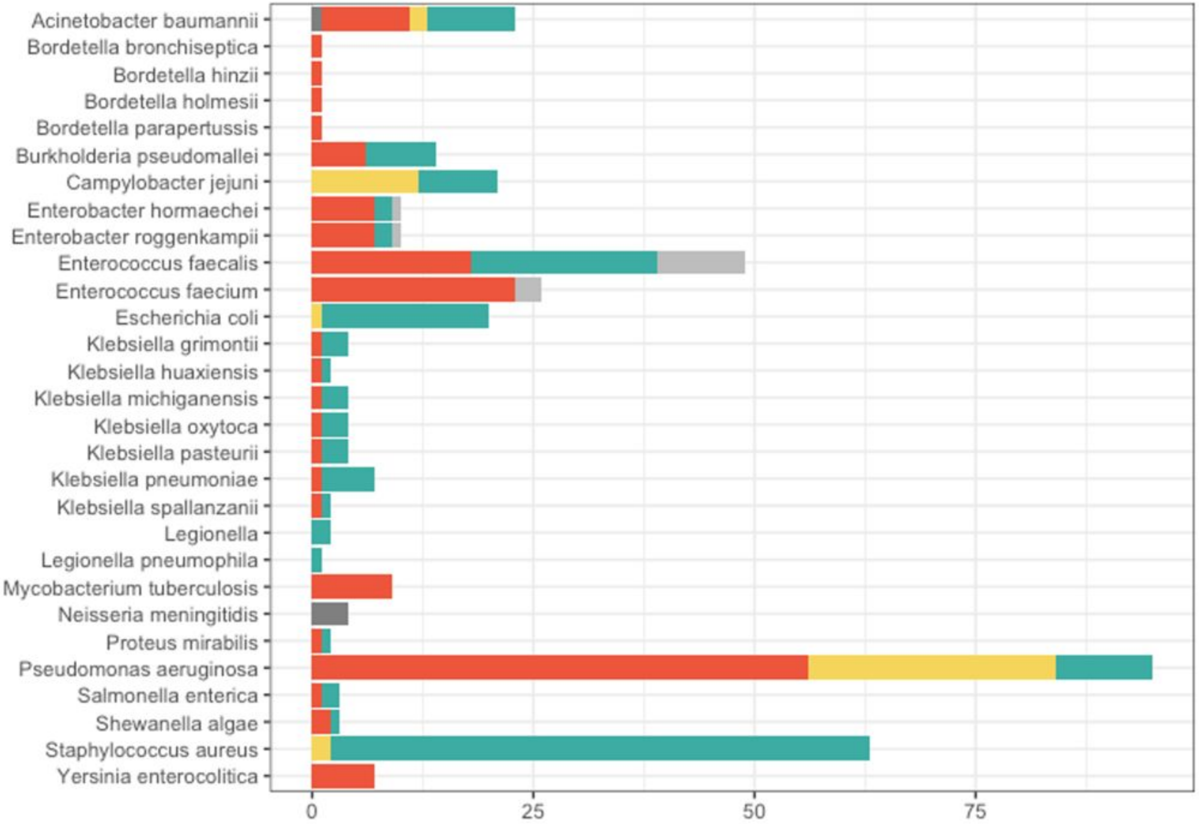
Rules to interpret *presence* of core genes



393 rules

WT S (43%)

WT I (11%)

WT R (40%)

135 unique genes

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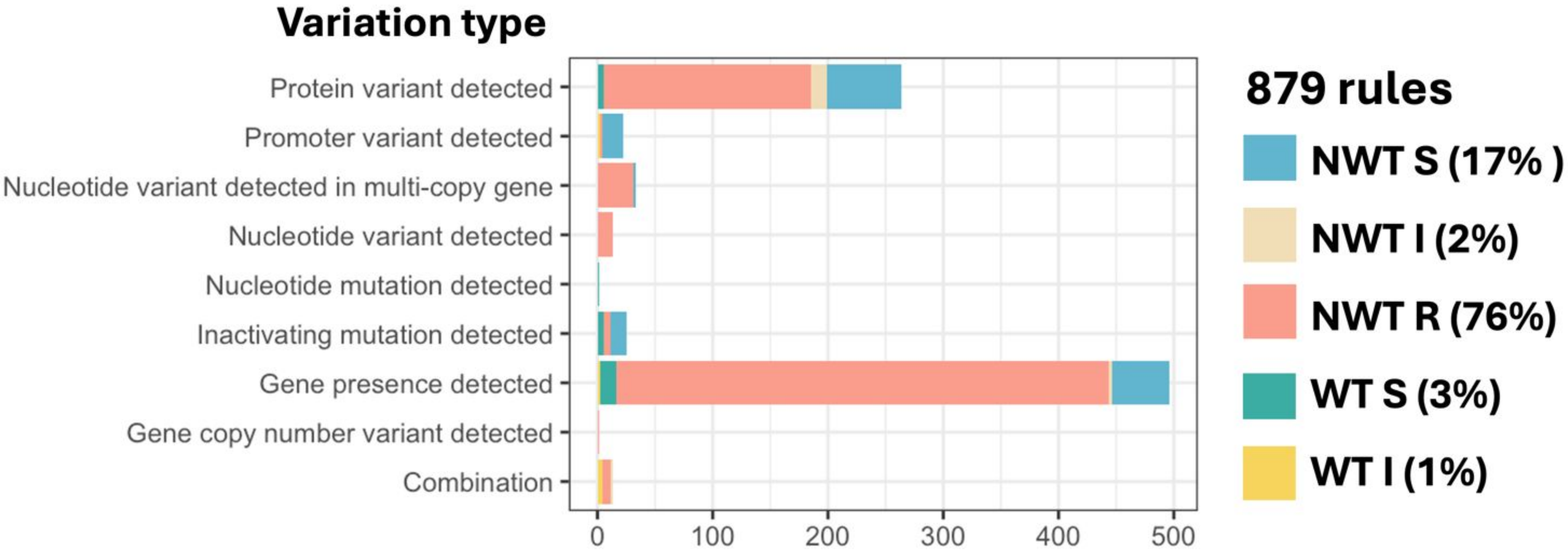
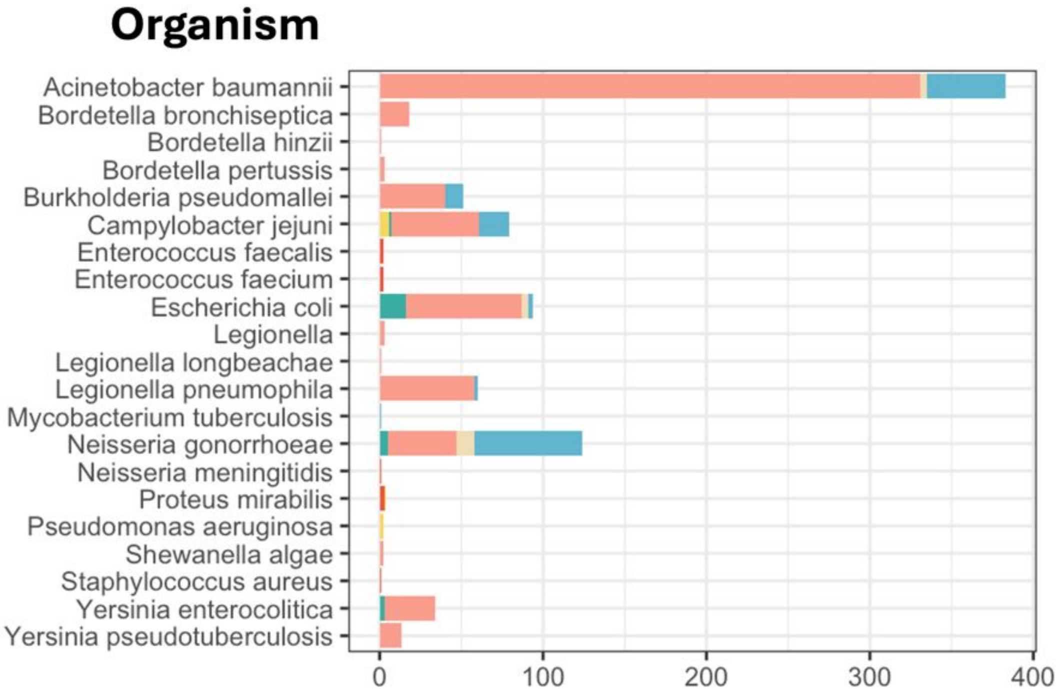
Rules for acquired variants

(i.e. rules other than detection of core gene presence)



Rules for acquired variants

(i.e. rules other than detection of core gene presence)



879 rules

- NWT S (17%)
- NWT I (2%)
- NWT R (76%)
- WT S (3%)
- WT I (1%)

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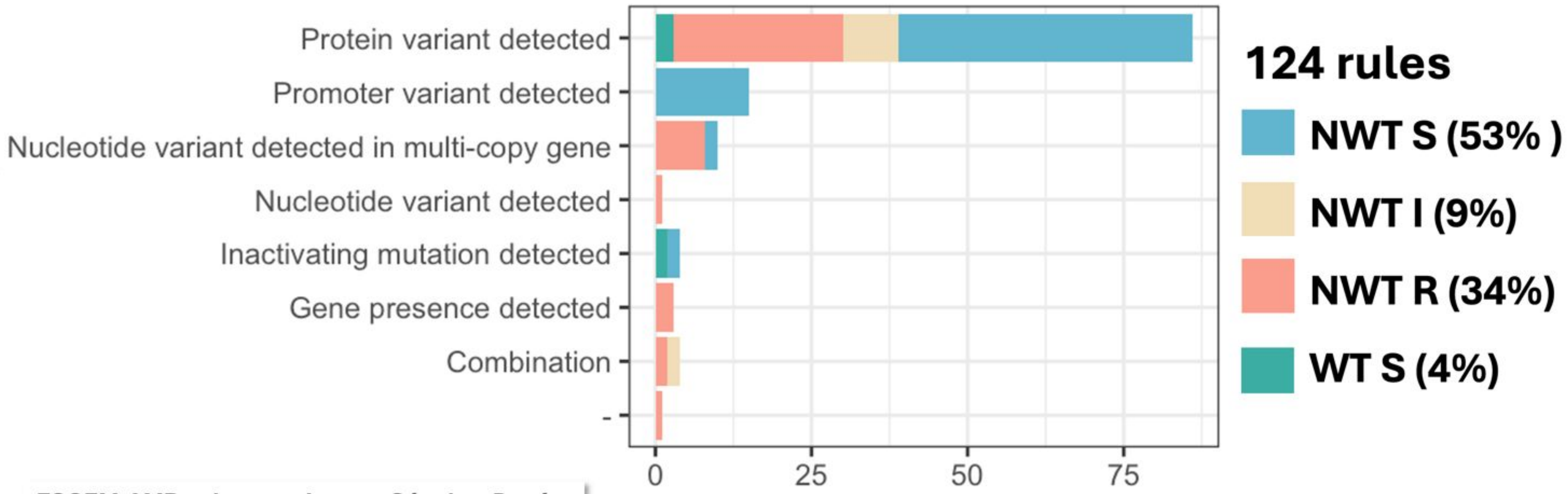
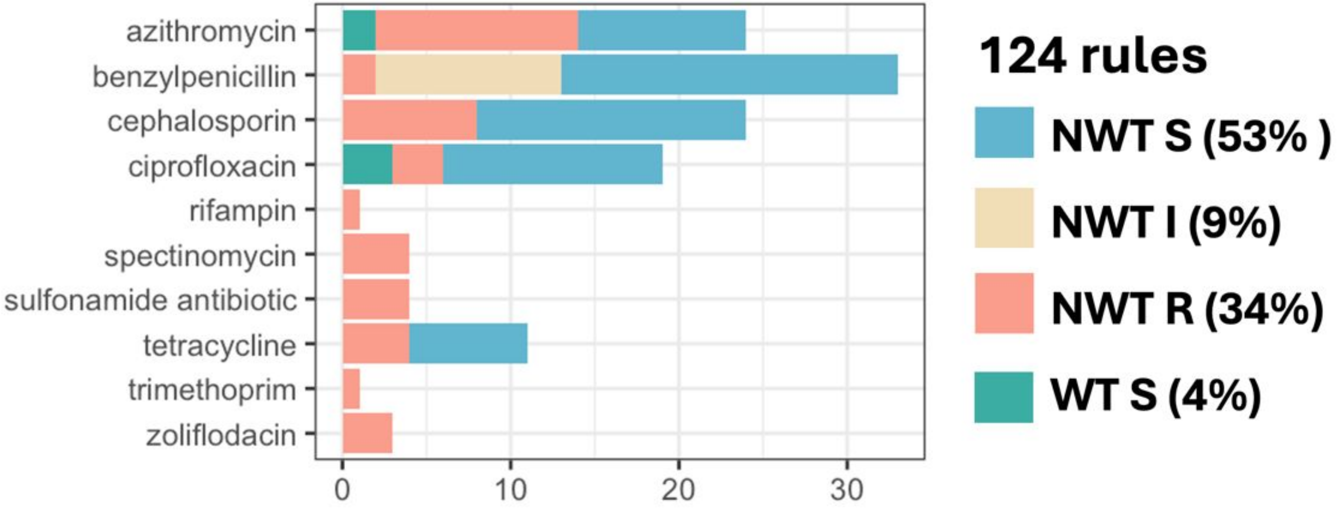
Acquired variants in *N. gonorrhoeae*



ESGEM-AMR subgroup: Leonor Sánchez Busó, Yonatan Grad, Sheeba Manoharan-Basil, Martin McHugh, Tatum Mortimer, Anna Roditscheff, Faina Wehrli, Adam Witney, Raffael Frei, Daniel Golparian, Magnus Unemo

Acquired variants in *N. gonorrhoeae*



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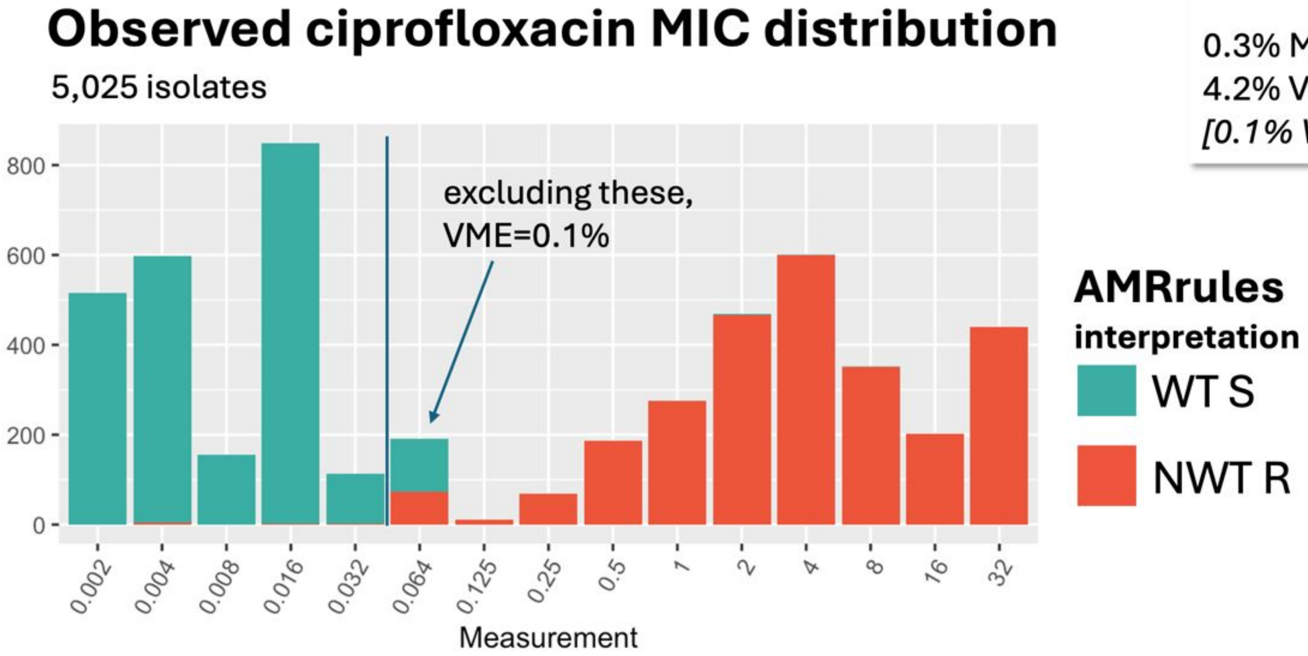
AMRrules interpretations for CIP in *N. gono*

Public data from EURO-GASP

Categorical concordance
95.7% Sensitivity
99.6% Specificity

0.3% Major error (S predicted R)
4.2% Very major error (R predicted S)
[0.1% VME excluding MIC=0.64]



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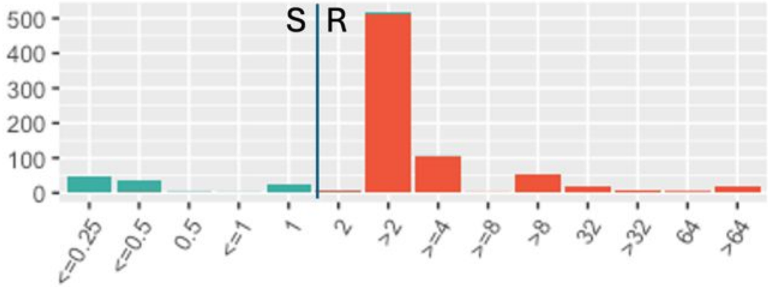
AMRrules interpretations for *A. baumannii*

Public data from EBI/CABBAGE

AMRrules interpretation

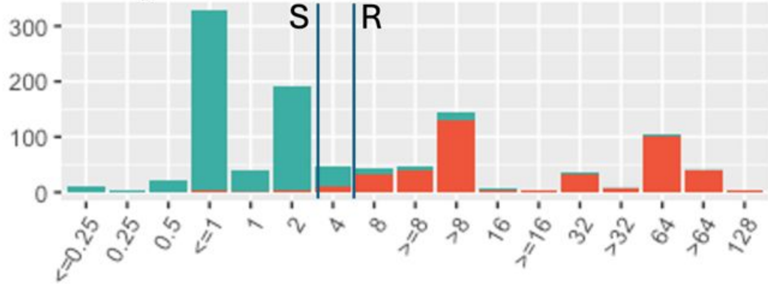
- WT S
- NWT R

Ciprofloxacin MIC (n=852)



Categorical concordance
98.1% Sensitivity
99.1% Specificity

Imipenem MIC (n=1,067)



Categorical concordance
90.6% Sensitivity
97.0% Specificity

ESGEM-AMR subgroup:
Paul Higgins, Bogdan Iorga, Rahul Garg, Mehrad Hamidian, Priyanka Khopkar-Kale, Margaret Lam, Bruno Silvester Lopes, Ignasi Roca, Varun Shamanna, Clement Tsui, David Wareham, Valeria Bortolaia, Adrian Egli, Lucie Amoureux, Vera Manageiro, Antoine Abou Fayad, Beverly Egyir

Links

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ESGEM-AMR Menu ≡

This is the home of the **ESGEM-AMR Working Group**, which focuses on curating organism-specific rule sets for interpreting AMR genotypes under the AMRrules framework.

The goal of AMRrules is to develop interpretive standards for AMR genotypes, akin to the interpretive standards developed by EUCAST and CLSI for antimicrobial susceptibility phenotyping.

An overview of the AMRrules concept and software is available in these slides and the AMRrules documentation, and version 1 rules curated by ESGEM-AMR are available here.

About ESGEM-AMR

In 2024 ESGEM, the ESCMID Study Group on Epidemiological Markers, partnered with the AMRrules team to form the ESGEM-AMR Working Group to curate organism-specific rule sets for AMRrules.

The convenors of ESGEM-AMR are Kat Holt (LSHTM), Natacha Couto (ESGEM Chair), and Jane Hawkey (Monash, leading bioinformatics development). We have ~200 members (listed here), structured into:

- Organism-focused subgroups who curate AMRrules for their focus organism (species or genus), and
- Data & Tools subgroup who developed the AMRrules specification and contribute to technical development and interoperability with other tools in the AMR genomics ecosystem.

https://esgem-amr.amrrules.org/

Home

Members

Resources

ESGEM-AMR Menu ≡

Member Resources

- AMRrules Specification & Syntax - v1.0 [docs]
- AMRrules Technical Guidance - v1.6 [PDF]
- AMRrules Browser [website]
- AMRrules Software pacakge [docs]
- AMRrules Documentation [docs]
- AMRgen R package for analysing matched genotype/phenotype data [AMRgen repo]
- AMRrulemakeR R package for automating rule generation using AMRgen analyses [AMRrulemakeR repo]
- Code and tools for accessing public AMRFinderPlus + AST data [datacuration repo]
- CARD/Antimicrobial Resistance Ontology (ARO): [browser]
- Variant specification: HGVS
- AMRrules syntax: [docs]
- EUCAST: [Breakpoints] [Expected Resistance]
- NCBI AMRFinderPlus: [software]
- NCBI refgene (AMR Gene Catalog): [browser]
- NCBI AMR Reference Gene Hierarchy: [browser] [TXT file download]
- NCBI refseq: [browser]
- NCBI AST data: [browser]
- NCBI AST data submission: [info] [submission format]
- NCBI Taxonomy: [browser]
- Meeting slides: [github]

Automatic validation of rules

Before submitting rules for review, they should be first checked using the Python rulevalidator.

ESGEM-AMR Menu ≡

Members by subgroup

Data & Tools

Jane Hawkey/Kat Holt, Andrew McArthur, Finlay Maguire, Michael Feldgarden, Brody Duncan, Kristy Horan, Leonid Chindelevitch, Kara Tsang, Amogelang Raphenya, Dag Harmsen, Emily Bordeleau, Mackenzie Wilke, Romain Pogorelcnik, Yu Wan, Zoe Dyson, Bogdan Iorga, Derek Sarovich, John Rossen, Silvia Argimon, Charlene Rodrigues, Rolf Kaas, Nick Duggett, Louise Teixeira Cerdeira, Matthijs Berends, Adrian Egli, João Perdigão, Tiffany Ta, Karyn Mukiri, Chiara Crestani, Jalees Nasir, Arjun Prasad

Enterococcus

Francesc Coll, Thomas Demuyser, Ana R. Freitas, Guido Werner, Precious Osadebamwen, Theo Goulliouris, Fiona Walsh, Valeria Bortolaia, Basil Britto Xavier, Helena Seth-Smith

Staphylococcus

Natacha Couto, Birgitta Duim, Valeria Bortolaia, Sarah Baines, Sandra Reuter, Assaf Rokney, Holly Grace Espiriu, Manal AbuOun, Sankarganesh Jeyaraj, Robert Kozak, Nick Duggett, Birgit Strommenger, Lina Cavaco, Varun Shamanna, Sabrina Di Gregorio, Teresa Ribeiro, Tim Read, Georgina Lewis-Woodhouse, Raphael Sieber

Acinetobacter baumannii

Paul Higgins, Rahul Garg, Mehrad Hamidian, Bogdan Iorga, Priyanka Khopkar-Kale, Margaret Lam, Bruno Silvester Lopes, Ignasi Roca, Varun Shamanna, Clement Tsui, David Wareham, Valeria Bortolaia, Adrian Egli, Lucie Amoureux, Vera Manageiro, Antoine Abou Fayad, Beverly Egyir

AMRrules Spec v1.1 - updates & CARD harmonisation

- New fields
  - Resistance mechanism
  - Gene family

- Updated fields
  - ARO accession
  - Variation type

AMRrules

New field: ‘resistance mechanism’

mechanism of antibiotic resistance

Download Sequences

https://card.mcmaster.ca/ontology/36007

Ontology	CARD's Antibiotic Resistance Ontology
Accession	ARO:1000002
Definition	Antibiotic resistance mechanisms evolve naturally via natural selection through random mutation, but it could also be engineered by applying an evolutionary stress on a population.
Parent Term(s)	**1 ontology terms** | Show
Sub-Term(s)	**10 ontology terms** | Hide + antibiotic target alteration [Resistance Mechanism] + antibiotic target replacement [Resistance Mechanism] + antibiotic target protection [Resistance Mechanism] + antibiotic inactivation [Resistance Mechanism] + antibiotic efflux [Resistance Mechanism] + reduced permeability to antibiotic [Resistance Mechanism] + resistance by absence [Resistance Mechanism] + modification to cell morphology [Resistance Mechanism] + resistance by host-dependent nutrient acquisition [Resistance Mechanism] + antibiotic target overexpression [Resistance Mechanism]

New field: ‘resistance mechanism’

Rule specification

ruleID	organism	gene	nodeID	variation type	gene context	drug	drug class	resistance mechanism	phenotype	clinical category
NMN0002	s__Neisseria meningitidis	farB	farB	Gene presence detected	core	-	-	antibiotic efflux	-	-
NMN0003	s__Neisseria meningitidis	mtrC	mtrC	Gene presence detected	core	-	-	antibiotic efflux	-	-
NMN0004	s__Neisseria meningitidis	norM	-	Gene presence detected	core	-	-	antibiotic efflux	-	-
NMN0005	s__Neisseria meningitidis	mtrR	mtrR	Gene presence detected	core	-	-	antibiotic efflux	-	-
ACI0160	s__Acinetobacter baumannii	amvA	amvA	Gene presence detected	core	-	-	-	-	-

If the variant is associated with a generic resistance mechanism that is not specific to a particular drug or class, it is permissible to leave both 'drug' and 'drug class' as '-' and indicate the 'resistance mechanism' instead (e.g. 'antibiotic efflux', ' reduced permeability to antibiotic').
When the AMRrules engine encounters such rules (i.e. with a mechanism but no drug or class), the associated markers will be assigned to a row in the report labelled with the mechanism term, and will not appear in any drug or class row.

Genome summary output

drug	drug class	clinical category	phenotype	evidence grade	markers (non-S)	markers (no rule)	markers (S)	ruleIDs	combo rules	organism
-	antibiotic efflux	-	-	-	-	farB, mtrC, norM, mtrR	-	-	-	s__Neisseria meningitidis

New field: 'gene family'

optional **field**
definition: name of gene family {ARO term}
we will autopopulate from CARD ARO index based on mapping of existing node/accession information

ruleID	gene	nodeID	protein accession	HMM accession	ARO accession	gene family	mutation	variation type	gene context
ACI0001	blaOXA-23_fam	blaOXA-23_fam	-	NF000266.2	ARO:3007710	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0002	blaOXA-23	blaOXA-23	WP_001046004.1	-	ARO:3001418	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0003	blaOXA-27	blaOXA-27	WP_063862443.1	-	ARO:3001422	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0004	blaOXA-49	blaOXA-49	WP_063864111.1	-	ARO:3001671	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0005	blaOXA-146	blaOXA-146	WP_063861044.1	-	ARO:3001779	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0006	blaOXA-165	blaOXA-165	WP_063861123.1	-	ARO:3001465	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0007	blaOXA-166	blaOXA-166	WP_063861129.1	-	ARO:3001466	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0030	blaOXA-24	blaOXA-24	WP_012754353.1	-	ARO:3001419	OXA-24-like beta-lactamase	-	Gene presence detected	acquired
ACI0031	blaOXA-72	blaOXA-72	WP_000713530.1	-	ARO:3001705	OXA-24-like beta-lactamase	-	Gene presence detected	acquired
ACI0032	blaOXA-58_fam	blaOXA-58_fam	-	NF000500.2	ARO:3007728	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0033	blaOXA-58	blaOXA-58	WP_002002480.1	-	ARO:3001611	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0034	blaOXA-96	blaOXA-96	WP_063864543.1	-	ARO:3001631	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0035	blaOXA-97	blaOXA-97	WP_063864544.1	-	ARO:3001647	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0036	blaOXA-164	blaOXA-164	WP_063861121.1	-	ARO:3001662	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0146	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	-	Gene presence detected	core
ACI0147	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	c.-7ins[ISAba1:inv]	Promoter variant detected	core
ACI0148	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	p.Leu167Val	Protein variant detected	core

Example: *Acinetobacter baumannii*

Updated field: 'ARO accession'

***required* field** (was previously optional)
definition: Antibiotic Resistance Ontology (ARO) identifier for the name of the gene (or gene family) this rule applies to
we will autopopulate based on mapping of existing node/accession information

ruleID	gene	nodeID	protein accession	HMM accession	ARO accession	gene family	mutation	variation type	gene context
ACI0001	blaOXA-23_fam	blaOXA-23_fam	-	NF000266.2	ARO:3007710	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0002	blaOXA-23	blaOXA-23	WP_001046004.1	-	ARO:3001418	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0003	blaOXA-27	blaOXA-27	WP_063862443.1	-	ARO:3001422	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0004	blaOXA-49	blaOXA-49	WP_063864111.1	-	ARO:3001671	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0005	blaOXA-146	blaOXA-146	WP_063861044.1	-	ARO:3001779	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0006	blaOXA-165	blaOXA-165	WP_063861123.1	-	ARO:3001465	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0007	blaOXA-166	blaOXA-166	WP_063861129.1	-	ARO:3001466	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0030	blaOXA-24	blaOXA-24	WP_012754353.1	-	ARO:3001419	OXA-24-like beta-lactamase	-	Gene presence detected	acquired
ACI0031	blaOXA-72	blaOXA-72	WP_000713530.1	-	ARO:3001705	OXA-24-like beta-lactamase	-	Gene presence detected	acquired
ACI0032	blaOXA-58_fam	blaOXA-58_fam	-	NF000500.2	ARO:3007728	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0033	blaOXA-58	blaOXA-58	WP_002002480.1	-	ARO:3001611	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0034	blaOXA-96	blaOXA-96	WP_063864543.1	-	ARO:3001631	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0035	blaOXA-97	blaOXA-97	WP_063864544.1	-	ARO:3001647	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0036	blaOXA-164	blaOXA-164	WP_063861121.1	-	ARO:3001662	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0146	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	-	Gene presence detected	core
ACI0147	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	c.-7ins[ISAba1:inv]	Promoter variant detected	core
ACI0148	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	p.Leu167Val	Protein variant detected	core

Example: *Acinetobacter baumannii*

Updated field: ‘variation type’

***required* field** (as before)
definition: explanation of the type of variation this rule applies to **{ARO term}**
These terms have now been added to CARD Antbiotic Resistance Ontology (ARO) so we have a formal controlled vocabulary that can be easily re-used in other tools

ruleID	gene	nodeID	protein accession	HMM accession	ARO accession	gene family	mutation	variation type	gene context
ACI0001	blaOXA-23_fam	blaOXA-23_fam	-	NF000266.2	ARO:3007710	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0002	blaOXA-23	blaOXA-23	WP_001046004.1	-	ARO:3001418	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0003	blaOXA-27	blaOXA-27	WP_063862443.1	-	ARO:3001422	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0004	blaOXA-49	blaOXA-49	WP_063864111.1	-	ARO:3001671	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0005	blaOXA-146	blaOXA-146	WP_063861044.1	-	ARO:3001779	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0006	blaOXA-165	blaOXA-165	WP_063861123.1	-	ARO:3001465	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0007	blaOXA-166	blaOXA-166	WP_063861129.1	-	ARO:3001466	OXA-23-like beta-lactamase	-	Gene presence detected	acquired
ACI0030	blaOXA-24	blaOXA-24	WP_012754353.1	-	ARO:3001419	OXA-24-like beta-lactamase	-	Gene presence detected	acquired
ACI0031	blaOXA-72	blaOXA-72	WP_000713530.1	-	ARO:3001705	OXA-24-like beta-lactamase	-	Gene presence detected	acquired
ACI0032	blaOXA-58_fam	blaOXA-58_fam	-	NF000500.2	ARO:3007728	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0033	blaOXA-58	blaOXA-58	WP_002002480.1	-	ARO:3001611	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0034	blaOXA-96	blaOXA-96	WP_063864543.1	-	ARO:3001631	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0035	blaOXA-97	blaOXA-97	WP_063864544.1	-	ARO:3001647	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0036	blaOXA-164	blaOXA-164	WP_063861121.1	-	ARO:3001662	OXA-58-like beta-lactamase	-	Gene presence detected	acquired
ACI0146	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	-	Gene presence detected	core
ACI0147	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	c.-7ins[ISAba1:inv]	Promoter variant detected	core
ACI0148	blaOXA-51_fam	blaOXA-51_fam	-	NF000268.2	ARO:3007725	OXA-51-like beta-lactamase	p.Leu167Val	Protein variant detected	core

Example: *Acinetobacter baumannii*

Values allowed in variation type column	The specified AMRrule applies if...	Notes or source
Gene presence detected	...the gene specified in the 'gene' column is detected as being present.	hAMRonization
Protein variant detected	...the protein variant specified in the 'mutation' column is detected in the specified 'gene'.	hAMRonization
Nucleotide variant detected	...the nucleotide variant specified in the 'mutation' column is detected in the specified 'gene'.	hAMRonization
Promoter variant detected	...the promoter variant specified in the 'mutation' column is detected in the specified 'gene'.	NCIT:C190205
Inactivating mutation detected	...the gene specified in the 'gene' column is inactivated by any type of mechanism (e.g. frameshift, internal stop, deletion, truncation), in the amino acid range specified in the 'mutation' column (or anywhere in the gene, if the 'mutation' column is blank i.e. '-').	NCIT:C178119
Gene truncation detected	...the gene specified in the 'gene' column is truncated, within the amino acid range specified in the 'mutation' column.	

https://docs.amrrules.org/en/latest/specification.html#variation-type

ARO Accession	ARO Name	ARO Defintion
ARO:3009567	ARO variant type	Type of variation reported by resistance gene annotators or interpreters. Based on the schemes developed by hAMRonization (https://github.com/pha4ge/hAMRonization) and AMRRules (https://docs.amrrules.org), with additional terms from the NCIT ontology.
ARO:3009568	gene presence detected	The gene has been detected in the query.
ARO:3009569	protein variant detected	The gene has been detected in the query but encodes a protein variant.
ARO:3009570	nucleotide variant detected	The gene has been detected in the query but encodes a nucleotide variant.
NCIT:C190205	promoter variant detected	A promoter has been detected in the query, but the promoter includes a nucleotide variant. Paraphrased from the NCI Thesaurus.
NCIT:C178119	inactivating mutation detected	The gene (or protein) has been detected in the query but includes a loss of function mutation (e.g., frameshift, internal stop, deletion, truncation). Paraphrased from the NCI Thesaurus.
ARO:3009571	gene truncation detected	The gene has been detected in the query but is truncated relative to a reference sequence.
NCIT:C189957	gene copy number variant detected	The gene has been detected in the query more than once. Paraphrased from the NCI Thesaurus.
ARO:3009572	nucleotide variant detected in multi-copy gene	The gene has been detected in the query more than once and at least one copy encodes a nucleotide variant.
ARO:3009573	low frequency variant detected	The sequencing reads support a mixed population, of which a fraction encode a nucleotide variant.
ARO:3009574	combination of variant types	A combination variant types.

https://card.mcmaster.ca/ontology/48425

AMRrules v2: Data-driven development of rules for acquired resistance

- **Protocols in development for**
  - Using AMRrulemakeR package to propose rules from geno-pheno data
  - Manual expert review and curation of proposed rules
    It is important that all rules are manually reviewed and curated by experts who have knowledge of the organism, the drug, the resistance mechanisms, and relevant literature

- **AMRrulemakeR code to be finalised - need to review following AMRgen updates**
  - Data-rich groups (*E. coli, K. pneumoniae, S. enterica, N. gonorrhoeae, E. faecium, S. aureus*) testing
  - Other groups should work on sourcing data if possible (we can assist)
  - If insufficient geno-pheno data, focus on defining rules for all markers currently reported by AMRfp in AllTheBacteria project based on literature

Sourcing Geno/Pheno data

Public data

- **AMRgen** R package to download geno+pheno from EBI/NCBI
- **AllTheBacteria** commandline tool to download AMRFinderPlus, MLST, etc
- **ESGEM-AMR** github with public dataset for all ESGEM-AMR supported organisms (Jane working on this)

AllTheBacteria Command Line Interface

for downloading ATB data, including AMRFP results

AllTheBacteria

https://github.com/allthebacteria/atb-cli

- Available for macOS, Linux & Windows
- Must download the database on first use (~540Mb) as ATB currently not hosted
  - `atb fetch`

```
# Get all AMR gene hits for E. coli (high-quality genomes only, limit to first 100 samples)
atb amr --species "Escherichia coli" --hq-only --limit 100

# Filter by drug class within E. coli
atb amr --species "Escherichia coli" --hq-only --class "BETA-LACTAM"

# Wildcard gene search (all beta-lactamase genes) in E. coli
atb amr --species "Escherichia coli" --gene "bla%"

# Get resistance genes for a drug class, across multiple species
atb amr --species "Escherichia coli,Klebsiella pneumoniae" --class "BETA-LACTAM"

# Find a gene across ALL genera (no species filter needed, limit to first 100 samples)
atb amr --gene "blaCTX-M-15" --limit 100
```

AllTheBacteria AMRFP results update

- Includes all genomes in ATB v0.2, incremental release 2024-08 **and 2025-05** (additional 322,920 genomes, total ~2.7 million)
  - used AMRFP v4.2.5, database 2025-12-03.1
  - all genomes except 7 completed successfully
- Will be run annually on all ATB data
- Now have a reproducible Nextflow pipeline (https://github.com/jhawkey/atb-amrfp-nf)
  - Can also be used for running smaller subsets (e.g. single species) when required, if there is a key AMRFP update
- **Upload to ATB database in progress**
- GitHub repo for AMRFP + ATB analysis
  - https://github.com/AMRverse/AllTheBacteria_AMRrules
  - Will contain reproducible analysis for all organisms with a working group
    - core genes, gene frequencies
    - matched public phenotype with genotypes

Sourcing Geno/Pheno data

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Collating public+private data

- **AMRgen** R package to import and harmonise data from different AST instruments and formats

Developing guidance on exporting AST from instruments

Started with Vitek

Please let us know if you can help with examples and instructions for other instruments!

Web app to import data from different formats

output files for submission to NCBI or EBI

http://apps.amrverse.org/amrgenconverter/

AMRgen converter

Upload antibiogram file

Browse... No file selected

Input format

Sensititre

Interpret resistance categories?

- Interpret using EUCAST breakpoints
- Interpret using CLSI breakpoints
- Interpret WT/NWT using ECOFF

Import and Summarise

Instructions | Imported data | Summary | Exports

How to use this app

This app imports antibiogram data from a variety of input formats, and exports submission-ready files suitable for submission to NCBI or ENA (via EBI AMR portal) as BioSample data. It is powered by the AMRgen and AMR for R packages.

1. Upload your antibiogram file in a supported format. Click the 'Input format' selector to see the list of supported input formats.
2. If you would like to re-interpret the MIC or disk measurements in the input file using latest breakpoints, check the relevant boxes under "Interpret resistance categories?"
3. Optionally, you can specify any additional import arguments using the fields below the Import button. See the AMRgen import_pheno() documentation for further guidance.
4. Click 'Import and Summarise' to load the data.
5. Review the imported data and summary tables, using the tabs above to navigate. See the AMRgen summarise_pheno() documentation for help with interpreting the summary tables.
6. Open the Exports tab to convert the imported data to submission-ready formats. For further details of the exported file formats and options, see the AMRgen functions:
    - export_ncbi_ast()
    - export_ebi_ast()
7. Note that NCBI and ENA require valid BioSamples be included in the submission files. If needed you can upload a second file mapping sample identifiers from your input antibiogram file, to BioSample accessions, under the Exports tab before downloading. This file can be in any tabular format, you just need to indicate which column contains sample identifiers that match your input file, and which column contains the BioSample accessions to replace these with in the exported files.

2026 Planning - Manuscripts

1. **AMRgen R package** - *to be submitted this week*
2. **AMRrules** (Q2) - *to be submitted by the end of July*
   - You should expect a paper to review around June
3. **Core gene rules** (Q3)
   - Why interpreting core gene rules is important but challenging
   - Approach to defining rules
   - Summary of rules - comparison across pathogens, gaps in knowledge
4. **Quantitative approach and acquired gene rules** (Q4)

Open for submissions

Submission deadline
29 May 2026

Genome Biology is calling for submissions to our Collection on the biology of drug resistance and tolerance in bacteria, viruses and fungi.

The emergence of drug resistance, tolerance and persistence in pathogenic microorganisms is a pressing global health challenge. The alarming rise of multidrug-resistant pathogens threatens to render many existing treatments ineffective. Recent advances in genomic technologies, such as whole genome sequencing (including the latest long read sequencing), have provided new insights into the genetic and functional aspects of drug resistance. By identifying resistance mechanisms and their evolutionary trajectories, researchers can contribute to the design of novel therapeutic interventions, thereby enhancing our capacity to combat...

Show more

Genome Biology
https://link.springer.com/collections/bjhecaieha

Collection

Cracking the code of AMR: resistance genomics

Submission status
Open

Submission deadline
24 June 2026

Antimicrobial resistance (AMR) poses one of the most significant global health challenges of our time, endangering the effectiveness of antibiotics and other antimicrobial agents. As the prevalence of resistant pathogens continues to rise, the need for innovative solutions to combat AMR has never been more urgent. The advent of genomics has revolutionized our understanding of microbial biology, enabling researchers to elucidate the genetic mechanisms underpinning resistance and facilitating the development of novel therapeutic strategies. By harnessing genomic technologies, we can better track the evolution of resistance, identify potential outbreaks, and ultimately inform public health interventions. This Collection aims to consolidate cutting-edge research that unites genomics and AMR, fostering interdisciplinary collaboration and knowledge exchange at a time when the stakes are higher than ever. —

npj AMR
https://www.nature.com/collections/cfbjiegbcg

AMRrules

ESCMID | ESGEM-AMR

2027 ESCMID Educational session (2h)

AMR genotyping - from determinants to interpretation

Chairs

Title	Speaker	Alternative Speaker
Interpretive standards for AMR genotyping	Natacha/Kat/Jane	Natacha/Kat/Jane

Questions? / Any other business?

ESGEM-AMR

ESCMID

AMRrules

amrrules.org